Lecture 1

Statistics, Sampling & Estimation

October 10, 2024

2110203 COMP ENG MATH II (CEDT)

CHULA
COMPUTER
ΣNGINEERING

Course Structure

- **Contents:**
  - Sampling & Estimation
  - Null-hypothesis Statistical Testing (NHST)
  - A/B Testing
  - Multi-arm Bandits

- **Instructors:**
  - Ekapol, Nat

	พฤ.เช้า	พฤ.บ่าย
10 Oct	Sampling & Estimation	Intro, Signals, Transformation
17 Oct	Testing	Convolution
24 Oct	Testing	AB Testing
31 Oct	Bandits	Lab
7 Nov	- (น่าจะ)	**สอบ stat**

Introduction to Statistics

Probability vs Statistics

- **Probability**
  - Toss a fair coin. What's the probability of getting 2 heads in 3 tosses?
  - Rules and logics concerning probability calculation. One correct answer.
- **Statistics**
  - You toss a coin of unknown bias 100 times. You got 55 heads. Draw some conclusion from this data.
  - More like an art. Make decisions based on possibly different assumptions. No single correct answer.

- **Known process** vs **Inferring from the outcome**

Statistics use case

- **Experimental design**
  - Make sure the collected data is useful.
  - *"To consult a statistician after an experiment is finished is often merely to ask him to conduct a post-mortem examination. He can perhaps say what the experiment died of."*
- **Provide summary**
  - Summarize data from a poll or questionnaire.
- **To infer knowledge**
  - Use data to draw inferences about model parameters.

Required knowledge

- Calculus (Diff, Integral)
- Bayes rules
- Discrete random variable
- Continuous random variable
- Law of Large Number
- Central Limit Theorem
- Joint Distributions

เพิ่งเรียนมาหมดเลย จำได้มั้ย

Sampling and Estimation

Simulation and Sampling

- We can simulate the outcome of a model.
- **Sampling:** get a set of numbers which follow a specific distribution.

Sampling How-to

```
import numpy as np
rng = np.random.default_rng()
```

```
print(rng.random())
print(rng.integers(1, 5))
```

```
0.9845715089427806
2
```

- We have use random number generator before.

Random from U(0,1]

Random from discrete U(1,4)

```
print(rng.exponential(1))
print(rng.normal(1, 2))
print(rng.geometric(0.1))
```

```
0.941294991737421
-1.0699011715923663
2
```

- Most scientific libraries (including *NumPy* and *Scipy*) has built-in functions.

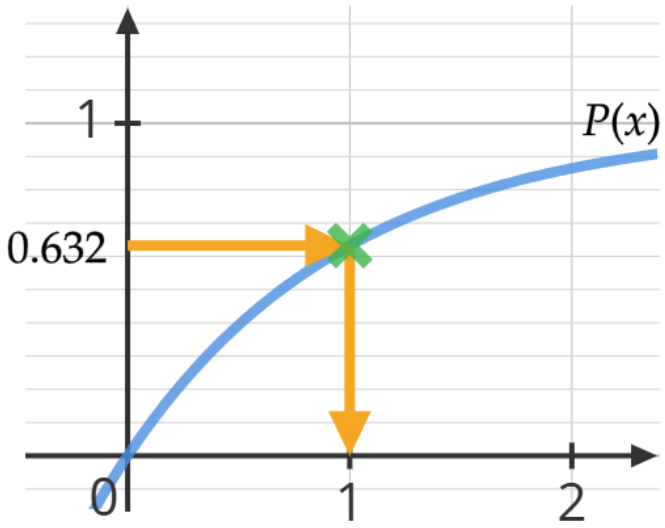
Sampling

- Random number generators follow **uniform distribution**.

- To sample from a specific distribution Y, we need a function:

$$f: x \rightarrow y \;\; (x \sim U(0,1), y \sim Y)$$

- **By definition, this is inverse CDF.**



Sampling Example

Problem. Let pdf $f_X(x) = 2x\ (0 \leq x \leq 1)$. Find $F_X^{-1}(x)$.

Answer.

First, find CDF $F_X(x) = \int_0^x 2x dx = \mathrm{x}^2\ (0 \leq \mathrm{x} \leq 1)$.

To find $F_X^{-1}(x)$:

$$F_X(x) = x^2$$

$$x = F_X^{-1}(x)^2$$

$$F_X^{-1}(x) = \sqrt{x}.$$

Estimation

- By **LLN** we know that we can estimate the mean by computing the average of large number of samples

- What about others?
  - Rate of decay in exponential
  - Variance of Gaussian

- Can we say how good is our estimate?

Estimation

- **Given the data, what are the mostly likely parameters?**

- **Example:** Toss a coin 100 times. The result is **55 HEADS** and **45 TAILS**. What is the **most likely p**, the chance of coin coming up with head in a single toss?

➔ **Attendance Check**

The probability of p

- We can formulate the problem as:

$$P(55\ heads) = \binom{100}{55} p^{55}(1-p)^{45} \qquad \textit{(binomial formula)}$$

- We should write this as:

$$P(55\ heads \boxed{|p)} = \binom{100}{55} p^{55}(1-p)^{45}$$

As this probability depend on p

"the probability of 55 heads given p"

Terminology

- **Experiment:** "Flip the coin 100 times and count the number of heads."
- **Data:** The result. "55 heads."
- **Parameter(s) of interest:** The unknown parameter p (usually denoted as θ)
- **Likelihood:** $P(55\ heads|p) = \binom{100}{55} p^{55}(1-p)^{45}$ or, more generally, $P(Data|p)$

Maximum Likelihood Estimation (MLE)

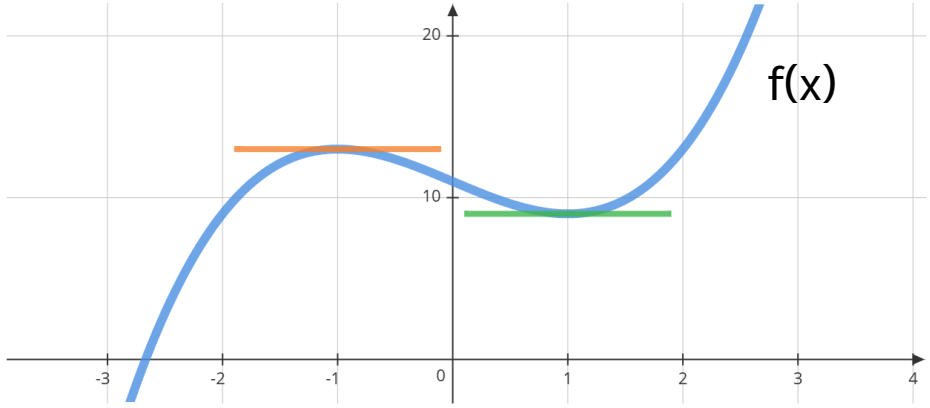
- **Maximum Likelihood Estimation (MLE)**

  → find p by **maximizing** the <u>likelihood function</u>.

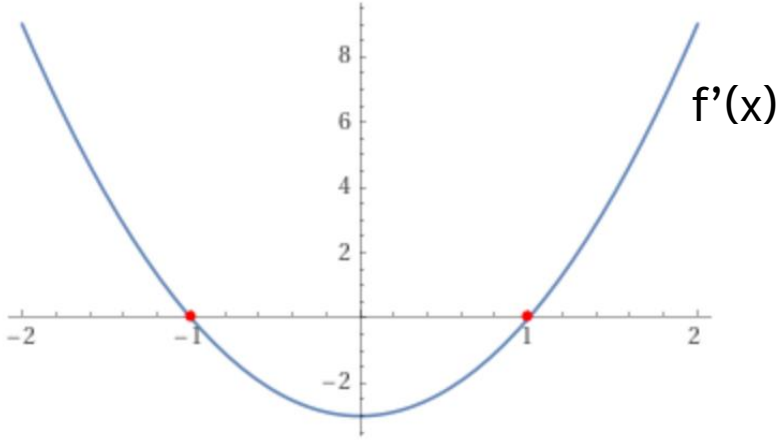
- We want to find: $\underset{p}{\operatorname{argmax}} \binom{100}{55} p^{55}(1-p)^{45}$

Interlude: Finding Extreme Point of Function

- Consider function $f(x) = x^3 - 3x + 11$.



- To find the **local** maxima and minima, we can do $f'(x) = 0$.



- To check whether it's maxima or minima, we check if $f''(x)$.
  If $f''(x) > 0$, it's minima. If $f''(x) < 0$, it's maxima.

MLE (cont'd)

We want to find: $\hat{p} = \underset{p}{\operatorname{argmax}} \binom{100}{55} p^{55}(1-p)^{45}$

$$\frac{d}{dp} P(Data|p) = \binom{100}{55} \underbrace{\left(p^{55}(-45)(1-p)^{44} + (1-p)^{45}(44)p^{54}\right)}_{\text{ดิฟผลคูณ}} = 0$$

Solve for p:

$$55p^{54}(1-p)^{45} = 45p^{55}(1-p)^{44}$$

$$55(1-p) = 45p$$

$$55 = 100p$$

Thus, MLE $\hat{p} = 0.55$

Log Likelihood

Using **natural log** of the likelihood function is often easier:

$$\ln P(55\ heads|p) = \underline{\ln\left(\binom{100}{55}\right) + 55\ln p + 45\ln(1-p)}$$

$\ln ab = \ln a + \ln b$

Since ln is an increasing function, the maxima is the same:

$$\frac{d}{dp}(\log likelihood) = \frac{d}{dp}\left[\ln\left(\binom{100}{55}\right) + 55\ln p + 45\ln(1-p)\right]$$

$$= \frac{55}{p} + \frac{45}{1-p} = 0$$

$$55(1-p) = 45p$$

Thus, MLE $\hat{p} = 0.55$

Example 1

Problem. Machines in Hamtaro's factory have their lifetime modelled by exponential distribution with an **unknown parameter.** Hamtaro found out that his machines failed after 2, 3, 1, 3, 4 years. Estimate the unknown parameter.

Answer. Let X_j be the RV that represent lifetime of machine i with actual value x_j. So pdf $f_{X_j}(x_i) = \lambda e^{-\lambda x_i}$. Thus:

$$f(x_1, x_2, x_3, x_4, x_5 | \lambda) = \lambda^5 e^{-\lambda(x_1+x_2+x_3+x_4+x_5)}.$$

Example 1 (sol.)

$$f(x_1, x_2, x_3, x_4, x_5|\lambda) = \lambda^5 e^{-\lambda(x_1+x_2+x_3+x_4+x_5)}.$$

Our data has value: 2, 3, 1, 3, 4, so we get the **log likelihood**:

$$\ln\big(f(2,3,1,3,4|\lambda)\big) = 5\ln\lambda - 13\lambda.$$

The MLE is thus: $\frac{d}{dx}(\log likelihood) = \frac{5}{\lambda} - 13 = 0$

Thus, $\hat{\lambda} = \frac{5}{13}$.

10-MIN BREAK

Likelihood Function

- **Discrete Random Variable:** the pmf.
  - **Binomial**: $P(n, k|\theta) = \binom{n}{k} k^{\theta}(1-k)^{\theta}$.
  - **Geometric:** $P(n|\theta) = (1-\theta)^{n}\theta$
- **Continuous Random Variable:** the pdf.
  - **Exponential:** $P(x|\theta) = -\theta e^{-\theta x}$
  - **Normal:** $P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Example 2

Problem. The number of attempts for the STATS COMP ENG class before passing following a Geometric distribution with unknown parameter. 3 students pass on their 1st, 1st, and 3rd try. Use MLE to find the unknown parameter.

Answer. The number of failure is 0, 0, 2. We get the likelihood function:

$$P(0,0,2|p) = (1-p)^{0+0+2}p^3$$

Using log likelihood: $\frac{\mathrm{d}}{\mathrm{dx}}[-2\ln(1-p) + 3\ln p] = 0$

$$-\frac{2}{1-p} + \frac{3}{p} = 0; \;\; \hat{p} = \frac{3}{5}.$$

MLE Properties

- Asymptotically unbiased
  - As the data grow, the estimated value get closer to real value.
- Asymptotically minimal variance
  - As data grow, MLE has the lowest variance among all unbiased estimators.

MLE in Nutshell

$$\underbrace{\boxed{P(\theta|D)}}_{\text{Posterior}} = \frac{\overbrace{\boxed{P(D|\theta)}}^{\text{Likelihood}}\overbrace{\boxed{P(\theta)}}^{\text{Prior}}}{\underbrace{\boxed{P(D)}}_{\text{Evidence}}}$$

In **MLE**, we maximize the likelihood $p(D|\theta)$.

- Assume Prior and Evidence is fixed (constant).
  - = assume the model is fixed.
- Done using log likelihood, taking partial derivative regarding θ and maximize the likelihood.

Maximum A Posteriori (MAP) Estimate

- Maximize the posterior (model parameter given data)

$$\underset{\theta}{\operatorname{argmax}}\ P(\theta|D) = \underset{\theta}{\operatorname{argmax}} \frac{P(D|\theta)P(\theta)}{P(D)}$$

- We can ignore $P(D)$ (as we take argsmax of θ)

- **Thus, $\underset{\boldsymbol{\theta}}{\boldsymbol{argmax}}\, \boldsymbol{P(D|\theta)P(\theta)}$.**
  - $P(D|\theta)$ is the likelihood.
  - $P(\theta)$ is the prior distribution of the parameter(s).

Example 3

- Toss an unknown coin 5 times. The result is [H, T, H, T, T]
- Koshi-kun told us that there are three possible coins
  - Coin A – P[H] = 0.8
  - Coin B – P[H] = 0.5
  - Coin C – P[H] = 0.4
- Koshi-kun also told us A, B, C has probability of being the coin with probability 0.4, 0.4, 0.2, respectively.
- Find the MAP estimate

Example 3 (sol)

- To find MAP estimate $\hat{p}_{MAP}$ we need to calculate for each prior:
  - **Coin A:** $p = 0.8, P(p = 0.8) = 0.4.$
    So, $P(p = 0.8|D) = P(D|p = 0.8)P(p = 0.8)/P(D).$
    $= C(5,2) \cdot 0.8^2(1 - 0.8)^3 \cdot 0.4/P(D) = \mathbf{0.02048}/P(D).$
  - **Coin B:** $p = 0.5, P(p = 0.5) = 0.4.$
    So, $P(p = 0.5|D) = P(D|p = 0.5)P(p = 0.5)/P(D).$
    $= C(5,2) \cdot 0.5^2(1 - 0.5)^3 \cdot 0.4/P(D) = \mathbf{0.125}/P(D).$
  - **Coin C:** $p = 0.4, P(p = 0.4) = 0.2.$
    So, $P(p = 0.4|D) = P(D|p = 0.4)P(p = 0.4)/P(D).$
    $= C(5,2) \cdot 0.4^2(1 - 0.6)^3 \cdot 0.2/P(D) = \mathbf{0.06912}/P(D).$
- Thus, $\hat{p}_{MAP} = 0.5$ as Coin B has **the maximum posterior.**

$$P(A|D) + P(B|D) + P(C|D) = 1$$
$$\therefore P(D) = 0.02048 + 0.125 + 0.06912 = 0.2146$$

$$P(A \mid D) = 0.02048/0.2146 = 0.0954$$
$$P(B \mid D) = 0.12500/0.2146 = 0.5825$$
$$P(C \mid D) = 0.06912/0.2146 = 0.3221$$

Example 4

Problem. What are the probability of getting the HEAD out of this coin?

Answer. $P(H|D) = \sum P(H, Coin|D) = \sum P(H|Coin, D)P(Coin|D)$

$= \sum P(H|Coin)P(Coin|D) = (0.8 \cdot 0.0954) + (0.5 \cdot 0.5825) + (0.4 \cdot 0.3221)$

$= \mathbf{0.49641} \qquad (\neq \mathbf{0.4} = \widehat{\boldsymbol{p}}_{\boldsymbol{MLE}})$

Example 5

Problem. Change the prior for coin A, B, C to 1, 1, 0, respectively. Find the MAP estimate.

Answer. $\hat{p}_{MAP} = 0.8$. The prior probability said that the only possible coin is coin A with P[H]=0.8.

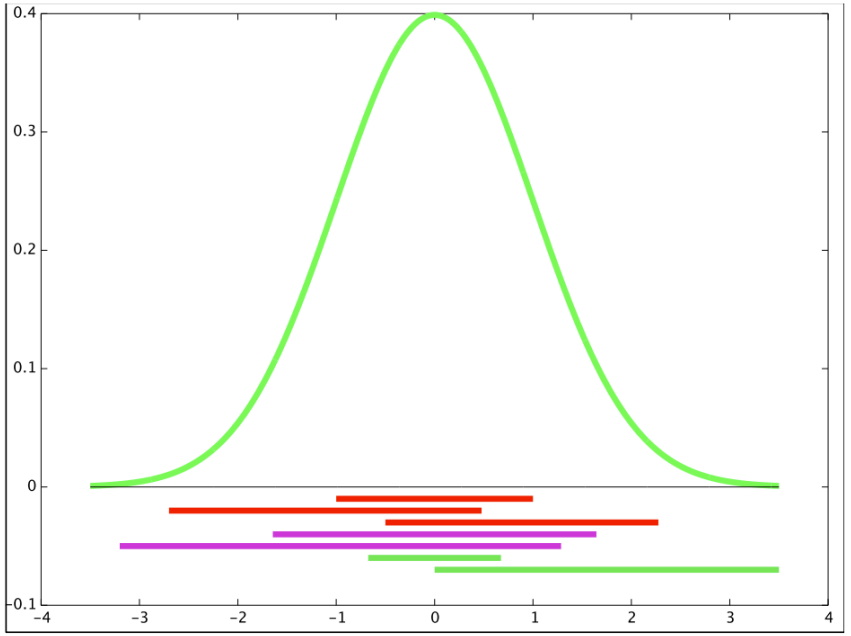
MAP Note

- A prior probability can radically change the result! (see previous example).

- Iterate updating posterior → prior → posterior
  - Computationally more expensive than MLE.

Probability interval

- A probability interval is some $[a, b]$ where $P(a \leq x \leq b) = p$.
  - Also called credible interval.
- NOT **confidence interval.**
- There can be multiple probability interval.



red = 0.68, magenta = 0.9, green = 0.5

Frequentist vs Bayesian

Frequentist

- Fixed model, random data.
- Maximize the **likelihood** (probability of data given a model)
- Does not care about prior probability.

Bayesian

- Fixed data, unknown model.
- Maximize the **posterior** (model parameters given data)
- Require a **prior** probability.
  - Unless completely known, each people can have different a priori belief.

Frequentist vs Bayesian

Probability (mathematics)

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$

Everyone uses Bayes' formula when the prior $P(H)$ is known.

Bayesian path

Frequentist path

Statistics (art)

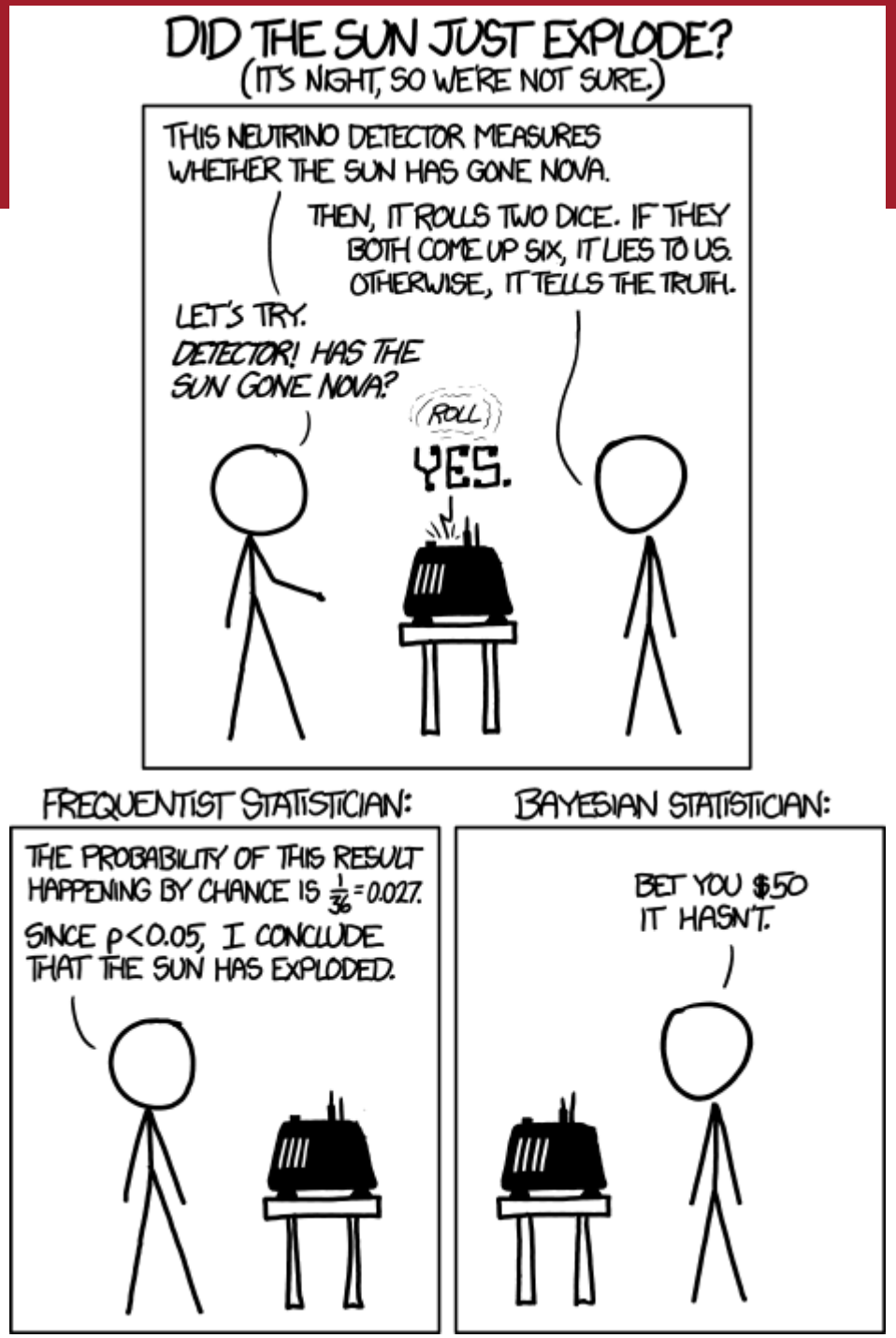
$$P_{\text{Posterior}}(H|D) = \frac{P(D|H)P_{\text{prior}}(H)}{P(D)}$$

Bayesians require a prior, so they develop one from the best information they have.

Likelihood $L(H;D) = P(D|H)$

Without a known prior frequentists draw inferences from just the likelihood function.

Did the sun explode?



https://xkcd.com/1132/

Reading

https://dspace.mit.edu/bitstream/handle/1721.1/153490/18-05-spring-2014/contents/readings/index.htm

- 10a, 10b, optional 11-16.

Statistics: Bayesian Inference		
5	C10	10a: Introduction to Statistics (PDF) 10b: Maximum Likelihood Estimates (PDF)
6	C11	11: Bayesian Updating with Discrete Priors (PDF)
	C12	12a: Bayesian Updating: Probabilistic Prediction (PDF) 12b: Bayesian Updating: Odds (PDF)
7	C13	13a: Bayesian Updating with Continuous Priors (PDF) 13b: Notational Conventions (PDF)
	C14	14a: Beta Distributions (PDF) 14b: Bayesian Updating with Continuous Data (PDF)
8	C15	15a: Conjugate Priors: Beta and Normal (PDF) 15b: Choosing Priors (PDF)
	C16	16: Probability Intervals (PDF)

HOMEWORK

- **STATS-CEDT-HW1.ipynb**
  - Only ASSIGNMENT 1 & 2 will be graded.
  - The rest are for your practise.
- Print as PDF for submission.

- **DEADLINE:** Wed, October 16, 2024.